

Smart City Bike Sharing Analysis Report

Overview:

The smart city bike sharing dataset can be used to analyze station performance, bike utilization, and city-wise operational efficiency. The Dataset consists of **2,856** rows and **12** columns. The analysis focuses on understanding where bike-sharing stations are concentrated, how bikes are distributed, station utilization patterns, and opportunities for improving operational efficiency. The dashboard was developed using Power BI with data cleaning, data modeling using a star schema approach, DAX calculations, descriptive statistics, and interactive visualizations.

Tables:

Fact_Bike_Station: Number, Name, Latitude, longitude, bankin, Bonus, status, contractname, Bike stands, Available Bike stands, Available Bikes, Date, Time, Bike availability status, Empty slot, Bike Availability percentage, Status Group, Station Demand Level.

Dim_Date: Date, Year, Month, Month_num, Month_Year.

Dim_Location: Station_ID, Station_Name, Latitude, Longitude.

Key Insights:

- City-wise station analysis identifies cities with higher numbers of bike-sharing stations, showing areas with stronger bike-sharing infrastructure.
- The available bikes by city analysis highlights differences in bike availability across locations, indicating that some cities have greater bike accessibility compared to others.
- Operational status analysis shows that the majority of stations are active, reflecting good availability and service coverage.
- The bike capacity versus availability analysis shows the relationship between station capacity and currently available bikes shows whether station resources are being effectively utilized.
- Station demand analysis categorizes stations based on demand levels and helps identify differences in usage patterns across stations.
- Descriptive statistics such as average and median available bikes provide an understanding of typical station availability.
- The geographic distribution map shows the concentration of bike-sharing stations across cities and helps identify major operational areas.
- The Top 10 station analysis highlights stations with higher bike availability and helps identify important locations for monitoring and resource planning.

Recommendations:

- Implement demand-based bike redistribution by moving bikes from low-demand stations to high-demand stations.
- Increase bike supply in cities showing higher demand and frequent availability requirements.
- Use utilization trends and availability patterns for better scheduling of bike redistribution activities.

Conclusion:

The dashboard provides a complete overview of bike-sharing operations by analyzing station distribution, availability, capacity, demand, and operational status. The insights generated can support better decision-making and help optimize bike allocation across cities for improved service efficiency.